

Numpy Linear Algebra

LAB(4)

Part -1

1. Write a NumPy program to test whether none of the elements of a given array are zero.
2. Write a NumPy program to test if any of the elements of a given array are non-zero.
3. Write a NumPy program to test a given array element-wise for finiteness (not infinity or not a number).
4. Write a NumPy program to test elements-wise for positive or negative infinity.
5. Write a NumPy program to test element-wise for NaN of a given array.
6. Write a NumPy program to create an element-wise comparison (greater, greater_equal, less and less_equal) of two given arrays.
7. Write a NumPy program to create an element-wise comparison (equal, equal within a tolerance) of two given arrays.
8. Write a NumPy program to create an array of 10 zeros, 10 ones, and 10 fives.
9. Write a NumPy program to create an array of integers from 30 to 70.
10. Write a NumPy program to create an array of all even integers from 30 to 70.
11. Write a NumPy program to create a 3x3 identity matrix.
12. Write a NumPy program to generate a random number between 0 and 1.
13. Write a NumPy program to create a vector with values ranging from 15 to 55 and print all values except the first and last.
14. Write a NumPy program to create a 3X4 array and iterate over it.
15. Write a NumPy program to create a vector of length 10 with values evenly distributed between 5 and 50.
16. Write a NumPy program to create a vector with values from 0 to 20 and change the sign of the numbers in the range from 9 to 15.
17. Write a NumPy program to multiply the values of two given vectors.
18. Write a NumPy program to create a 3x4 matrix filled with values from 10 to 21.
19. Write a NumPy program to find the number of rows and columns in a given matrix.
20. Write a NumPy program to create a 5x5 zero matrix with elements on the main diagonal equal to 1, 2, 3, 4, 5.
21. Write a NumPy program to create a 3x3x3 array filled with arbitrary values.
22. Write a NumPy program to compute the sum of all elements, the sum of each column and the sum of each row in a given array.
23. Write a NumPy program to compute the inner product of two given vectors.
24. Write a NumPy program to add a vector to each row of a given matrix.
25. Write a NumPy program to check whether two arrays are equal (element wise) or not.

26. Write a NumPy program to create a new array of given shape (5,6) and type, filled with zeros.
27. Write a NumPy program to sort a given array by row and column in ascending order.
28. Write a NumPy program to extract all numbers from a given array less and greater than a specified number.
29. Write a NumPy program to replace all numbers in a given array equal, less and greater than a given number.
30. Write a NumPy program to create a 4x4 array. Create an array from said array by swapping first and last, second and third columns.
31. Write a NumPy program to swap rows and columns of a given array in reverse order.
32. Write a NumPy program to multiply two given arrays of the same size element-by-element.

Part-2

1. Write a NumPy program to convert a list of numeric values into a one-dimensional NumPy array.

Expected Output:

Original List: [12.23, 13.32, 100, 36.32]

One-dimensional NumPy array: [12.23 13.32 100. 36.32]

2. Write a NumPy program to create a 3x3 matrix with values ranging from 2 to 10.

Expected Output:

```
[[ 2  3  4]
```

```
 [ 5  6  7]
```

```
 [ 8  9 10]]
```

3. Write a NumPy program to create a null vector of size 10 and update the sixth value to 11.

```
[ 0.  0.  0.  0.  0.  0.  0.  0.  0.  0.]
```

Update sixth value to 11

```
[ 0.  0.  0.  0.  0.  0. 11.  0.  0.  0.]
```

4. Write a NumPy program to create an array with values ranging from 12 to 38.

Expected Output:

```
[12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34  
35 36 37]
```

5. Write a NumPy program to reverse an array (the first element becomes the last).

Original array:

```
[12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34  
35 36 37]
```

Reverse array:

```
[37 36 35 34 33 32 31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15  
14 13 12]
```

6. Write a NumPy program to convert an array to a floating type.

```
Sample output:
Original array
[1, 2, 3, 4]
Array converted to a float type:
[ 1.  2.  3.  4.]
```

7. Write a NumPy program to append values to the end of an array.

```
Expected Output:
Original array:
[10, 20, 30]
After append values to the end of the array:
[10 20 30 40 50 60 70 80 90]
```

8. Write a NumPy program to create an empty and full array.

```
Expected Output:
[ 6.93270651e-310  1.59262180e-316  6.93270559e-310  6.93270665e-310]
[ 6.93270667e-310  6.93270671e-310  6.93270668e-310  6.93270483e-310]
[ 6.93270668e-310  6.93270671e-310  6.93270370e-310  6.93270488e-310]]
[[6 6 6]
 [6 6 6]
 [6 6 6]]
```

9. Write a NumPy program to convert Centigrade degrees into Fahrenheit degrees.
Centigrade values are stored in a NumPy array.

```
Sample Array [0, 12, 45.21, 34, 99.91]
[-17.78, -11.11, 7.34, 1.11, 37.73, 0. ]
Expected Output:
Values in Fahrenheit degrees:
[ 0. 12. 45.21 34. 99.91 32. ]
Values in Centigrade degrees:
[-17.78 -11.11  7.34  1.11 37.73  0. ]
Values in Centigrade degrees:
[-17.78 -11.11  7.34  1.11 37.73  0. ]
Values in Fahrenheit degrees:
[-0. 12. 45.21 34. 99.91 32. ]
```

10. Write a NumPy program to test whether each element of a 1-D array is also present in a second array.

```
Expected Output:
Array1: [ 0 10 20 40 60]
Array2: [0, 40]
Compare each element of array1 and array2
[ True False False True False]
```

11. Write a NumPy program to find common values between two arrays.

```
Expected Output:  
Array1: [ 0 10 20 40 60]  
Array2: [10, 30, 40]  
Common values between two arrays:  
[10 40]
```

12. Write a NumPy program to get the unique elements of an array.

```
Expected Output:  
Original array:  
[10 10 20 20 30 30]  
Unique elements of the above array:  
[10 20 30]  
Original array:  
[[1 1]  
 [2 3]]  
Unique elements of the above array:  
[1 2 3]
```

13. Write a NumPy program to find the set difference between two arrays. The set difference will return sorted, distinct values in array1 that are not in array2.

```
Expected Output:  
Array1: [ 0 10 20 40 60 80]  
Array2: [10, 30, 40, 50, 70, 90]  
Set difference between two arrays:  
[ 0 20 60 80]
```

14. Write a NumPy program to compare two arrays using NumPy.

```
Array a: [1 2]  
Array b: [4 5]  
a > b  
[False False]  
a >= b  
[False False]  
a < b  
[ True True]  
a <= b  
[ True True]
```

15. Write a NumPy program to sort along the first and last axes of an array.

```
Sample array: [[2,5],[4,4]]
Expected Output:
Original array:
[[4 6]
 [2 1]]
Sort along the first axis:
[[2 1]
 [4 6]]
Sort along the last axis:
[[1 2]
 [4 6]]
```

16. Write a NumPy program to get the values and indices of the elements that are bigger than 10 in a given array.

```
Original array:
[[ 0 10 20]
 [20 30 40]]
Values bigger than 10 = [20 20 30 40]
Their indices are (array([0, 1, 1, 1]), array([2, 0, 1, 2]))
```

17. Write a NumPy program to create a contiguous flattened array.

```
Original array:
[[10 20 30]
 [20 40 50]]
New flattened array:
[10 20 30 20 40 50]
```

18. Write a NumPy program to create a 2-dimensional array of size 2 x 3 (composed of 4-byte integer elements), also print the shape, type and data type of the array.

```
Expected Output:
(2, 3)
```

19. Write a NumPy program to create another shape from an array without changing its data.

```
Reshape 3x2:
[[1 2]
 [3 4]
 [5 6]]
Reshape 2x3:
[[1 2 3]
 [4 5 6]]
```

20. Write a NumPy program to create a new array of 3*5, filled with 2.

Expected Output:

```
[[2 2 2 2 2]
 [2 2 2 2 2]
 [2 2 2 2 2]]
[[2 2 2 2 2]
 [2 2 2 2 2]
 [2 2 2 2 2]]
```

21. Write a NumPy program to create an array of 10's with the same shape and type as the given array.

Expected Output:

```
[10 10 10 10]
```

22. Write a NumPy program to create a 2-D array whose diagonal equals [4, 5, 6, 8] and 0's elsewhere.

Expected Output:

```
[[4 0 0 0]
 [0 5 0 0]
 [0 0 6 0]
 [0 0 0 8]]
```

23. Write a NumPy program to create a 1-D array with values from 0 to 50 and an array from 10 to 50.

Expected Output:

Array from 0 to 50:

```
[ 0  1  2  3  4  5  6  7  8  9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24
 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47
 48 49]
```

Array from 10 to 50:

```
[10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32
 33 34
 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49]
```

24. Write a NumPy program to find the 4th element of a specified array.

Expected Output:

```
[[ 2 4 6]
```

```
[ 6 8 10]]
```

Forth element of the array:

```
6
```

25. Write a NumPy program to test whether specified values are present in an array.

```
Expected Output:  
Original array:  
[[ 1.12  2.  3.45]  
 [ 2.33  5.12  6.  ]]  
True  
False  
True  
False  
True
```


